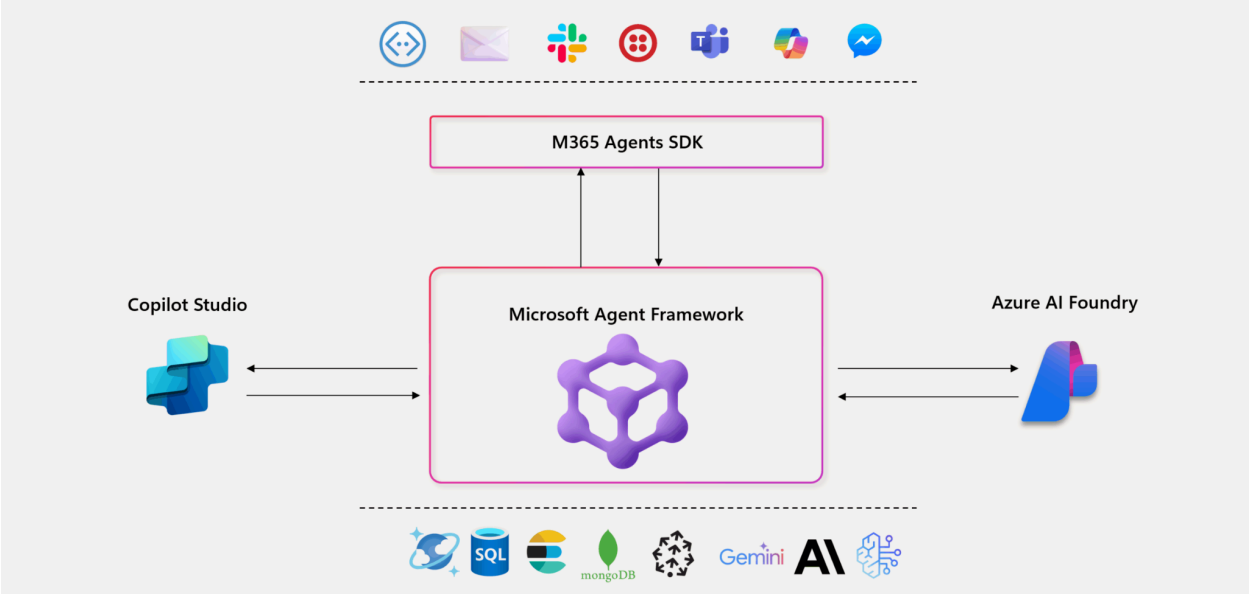
1. What is Microsoft Agent Framework (MAF)?

Microsoft Agent Framework (MAF) is an open-source development framework by Microsoft used to build, manage, and orchestrate AI agents and multi-agent systems.

It provides developers with the tools to create intelligent applications that can:

- Understand user instructions.
- Make decisions using Large Language Models (LLMs).
- Execute Python functions and external tools.
- Maintain conversation history and memory.
- Collaborate with other AI agents.
- Automate complex, multi-step tasks.

It is the successor to Microsoft's AutoGen and Semantic Kernel, combining their capabilities into one framework.



Key features of MAF

1. AI Agents

- Individual AI assistants that understand instructions.
- Use LLMs to process user requests.
- Can call tools and perform tasks autonomously.

2. Multi-Agent Collaboration

- Multiple AI agents work together to complete a task.
- Each agent can have a specific responsibility.
- Agents can share information and coordinate tasks.

3. Tool Integration

- Agents can execute external functions and APIs.
- Can connect to databases, search engines, and other services.
- Supports Model Context Protocol (MCP) integrations.

4. Memory Management

- Maintains conversation context.
- Supports multi-turn conversations.
- Helps agents remember relevant information across interactions.

5. Workflow Orchestration

- Controls how multiple agents and functions execute.
- Supports sequential and parallel execution.
- Supports conditional workflows and human approval.

6. Model Integration

- Supports different LLM providers.
- Can integrate with Azure OpenAI, OpenAI, Anthropic, Ollama, and Microsoft Foundry.

Microsoft Learn+1

Main components of MAF

Component	Purpose
Agents	Perform tasks using LLMs and tools.
Workflows	Coordinate multiple tasks and agents.
Memory	Maintain conversation context.

Tools	Allow agents to interact with external systems.
Middleware	Intercepts and controls agent operations.
Agent Harness	Provides reusable capabilities such as planning, memory, and tool management.

How does MAF work?

User Input

Example: Find me a laptop under ₹80,000.

🔽

AI Agent (MAF)

Understands the request and decides what to do.

🔽

Tool Execution

Calls product search API or database.

🔽

Process Results

Filters and compares available laptops.

🔽

Final Response

Recommends suitable laptops to the user.

Example applications

- AI customer support agents.
- Automated shopping assistants.

- Multi-agent research systems.
- HR recruitment assistants.
- Business process automation.
- AI-powered data analysis.

Technologies used with MAF

- Python
- C#
- Large Language Models (LLMs)
- Azure OpenAI
- Microsoft Foundry
- MCP (Model Context Protocol)
- APIs and external tools

What is Microsoft Foundry Agent Service?

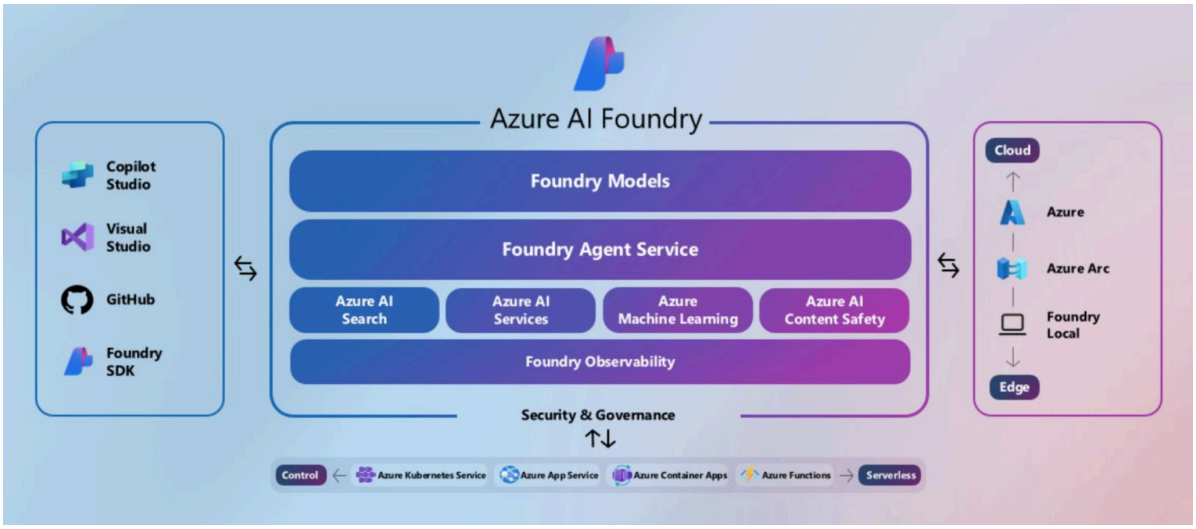
Microsoft Foundry Agent Service is a fully managed cloud service on Microsoft Azure that helps developers build, deploy, manage, and scale AI agents.

It provides the infrastructure required to run AI agents in real-world production environments.

In simple terms:

- MAF helps developers build the AI agent.
- Foundry Agent Service helps developers host, deploy, manage, and scale the AI agent.

It reduces the need to manage servers, infrastructure, authentication, and monitoring manually.



Key features of Foundry Agent Service

1. Agent Hosting

- Hosts AI agents in the cloud.
- Provides managed runtime environments.
- Allows agents to run without maintaining your own hosting infrastructure.

2. Model Integration

- Supports models available through the Microsoft Foundry model catalog.
- Allows developers to choose suitable models.
- Supports changing models without completely rebuilding the application.

3. Tool Integration
Agents can connect to:

- Web search.
- File search.
- Code Interpreter.
- MCP servers.
- Custom functions and APIs.

4. Security and Authentication

- Integrates with Microsoft Entra ID.
- Supports Role-Based Access Control (RBAC).
- Provides network isolation and security controls.
- Helps protect enterprise data.

5. Monitoring and Observability

- Tracks agent execution.
- Provides logs, metrics, and traces.
- Helps developers identify errors and evaluate agent performance.

6. Scalability

- Automatically scales agent hosting resources.
- Supports enterprise-level workloads.
- Handles increasing requests and usage.

7. Publishing and Sharing

- Supports agent versioning and publishing.
- Allows agents to be shared through Microsoft Teams and Microsoft 365 Copilot.

Types of agents in Foundry Agent Service

There are three important agent types in the current documentation.

Agent type	Description
Prompt Agents	Created by configuring instructions, models, and tools. Foundry manages execution.
Hosted Agents	Developers bring their own code and framework. Foundry hosts and manages execution.
Voice-based Prompt Agents	Support real-time voice interactions with managed orchestration.

C. Architecture of Foundry Agent Service

User / Application

Web application, mobile application, or Teams

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Microsoft Foundry Agent Service

Managed agent runtime and orchestration

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Models

GPT, Llama, etc.

🔽

Tools

Search, APIs, MCP

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Security

Identity and RBAC

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Monitoring

Logs and metrics

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Deployed AI Agent

Available for real-world users and applications

D. Real-world example

Consider an AI-powered customer support system.

1. A customer asks about an order.
2. The AI agent understands the question.
3. The agent connects to an order management API.
4. It retrieves the order details.
5. It generates a response.
6. Foundry Agent Service hosts the agent and manages its runtime, security, and monitoring.

Difference between MAF and Foundry Agent Service

Feature	Microsoft Agent Framework	Foundry Agent Service
Type	Development framework	Managed cloud service
Main purpose	Build AI agents	Deploy and manage AI agents
Environment	Local development or cloud hosting	Microsoft Azure
Programming	Python, C# and supported SDKs	Portal, SDKs, APIs and hosted code
Agent creation	Yes	Yes
Agent orchestration	Yes	Managed runtime and tools
Hosting	Can be self-hosted or integrated with hosting platforms	Managed hosting
Monitoring	Framework integrations and telemetry	Built-in observability
Security	Framework-level security capabilities	Enterprise identity and access controls